College of Engineering and Technology

Project Work/Internship – Student Hand Book

Project Batch ID
364

Name of student	Register Number	Department	Mobile Number	Email ID
Kavisha Arora	RA2211003010480	Ctech	9161687774	ka1189@srmist.edu.in
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Degree/ program	B.Tech	**Specialisation**	Computer Science and Engineering Specialisation with Core
Academic Year	2025-2026 (Even)	**Semester**	8
Course Code	21CSP401L/ 21CSP402L	**Course Title**	Project

Working Title of the Project:	AI-Driven Smart Energy Optimization and Carbon Analytics Platform .
Project Site / Location	
Name and address of the company/organisation (Applicable for projects with industry or industry support)	SRM IST, Kattankulathur, Chengalpattu District-603203

Supervision Team			
	Supervisor	**Co-Supervisor**	**External Supervisor (If applicable)**
Name			
Designation			
Department			
Campus	Kattankulathur	Kattankulathur	
Telephone			
E-mail			

Mission Statement

The proposed mission of the AI-Driven Smart Energy Optimisation and Carbon Analytics Platform involves designing and developing an intelligent, scalable, and data-driven system that can proactively facilitate energy optimisation and carbon impact analysis. The proposed project applies the concept of artificial intelligence to support efficient energy use while promoting sustainability-driven decision-making in institutional and commercial settings.

Problem (or) Product Description:

The consequent growth in energy use in institutional and commercial infrastructure has led to rises in operating expenses and environmental emissions of carbon dioxide. The traditional energy management solution is a reactive one that is focused on past data analysis and static reporting, but is not particularly concerned with predictive analytics or sustainable practices.

The AI-driven Smart Energy Optimisation and Carbon Analytics Platform bridges these gaps by combining machine learning energy forecasting models, anomaly detection analysis, and carbon emission modelling under a single analytical framework. This platform benefits users by allowing them to forecast future energy requirements and explore energy optimisation plans informed by carbon sensitivity.

Assumptions and Constraints

ASSUMPTIONS

- Energy consumption data is available in a structured digital format.
- Standard emission factors can be applied for carbon estimation.
- Users possess a basic understanding of energy analytics dashboards.

CONSTRAINTS

- Limited access to real-time IoT sensor data during initial deployment.
- Dependence on data quality for model accuracy.
- Computational limitations for advanced deep learning models.

Stakeholders

- Facility and Energy Managers
- Sustainability and ESG Teams
- Institutional Administrators
- Data Analysts and Researchers
- Academic Supervisors

Division of work and contributors

Time period		Activities or components of the project	Name/Register Number of the Individual Contributor	Names/Register Number of the Joint Contributors
From Date	To Date			
Nov’25	Dec’25	Database connection and schema design	Kavisha	Karan
Dec’25	Dec’25	Backend layer integration	Karan	Kavisha
Dec’25	Dec’25	Analytics layer implementation	Karan	Kavisha
Dec’25	Dec’25	ML model integration	Karan	Kavisha
Dec’25	Jan’25	Power BI dashboard integration	Kavisha	Karan
Jan’26	Jan '26	UI dashboard development (ongoing)	Kavisha	Karan
Jan '26	Feb '26	Anomaly detection engine implementation with residual Z-score scoring and source attribution	Kavisha	Karan
Jan '26	Feb '26	JWT + bcrypt authentication, RBAC implementation (Admin / User / Viewer roles)	Karan	Kavisha
Jan '26	Feb '26	Frontend dashboard — energy trends, anomaly insights, forecasting, carbon analytics tabs	Kavisha	Karan
Jan '26	Feb '26	Carbon analytics logic and emission estimation module integration	Karan	Kavisha
Jan '26	Feb '26	Centralized INR tariff utility and cost standardisation across all module	Karan	Kavisha
Jan '26	Feb '26	Profile Management	Karan	Kavisha

Summary record of major progress meetings with supervisors (Sprint 1)

Summary record of major progress meetings with supervisors		Database connection completed. All system layers successfully connected. Live Power BI dashboard demonstrated with sample energy data and analytics outputs. UI dashboard development to be completed. Model testing and validation to be initiated.	**Working title of dissertation/research project:** AI-Driven Smart Energy Optimisation and Carbon Analytics Platform.	
Meeting date & supervisors present	**Progress since last meeting**	**Agreed programme of work and target dates**	**Other issues, e.g. facilities, supervision, training needs, etc.**	**Date of next meeting**
2/1/26	Errors removed and continued building	Before Review 2		

Summary record of major progress meetings with supervisors (SPRINT 2)

<table>
<tr><td colspan="2">Summary record of major progress meetings with supervisors</td><td>Analytics and forecasting layers integrated with backend. Carbon analytics logic incorporated.

UI dashboard completion</td><td colspan="2">Working title of dissertation/research project:

AI-Driven Smart Energy Optimization and Carbon Analytics Platform .</td></tr>
<tr><td>Meeting date & supervisors present</td><td>Progress since last meeting</td><td>Agreed programme of work and target dates</td><td>Other issues, e.g. facilities, supervision, training needs, etc.</td><td>Date of next meeting</td></tr>
<tr><td>7/2/2026</td><td>Anomaly detection module corrected — residual Z-score scoring implemented. Backend query aggregation issues resolved. Baseline calculations validated.</td><td>Complete UI dashboard tabs and integrate carbon analytics before Review 2</td><td>Anomaly query bug required significant rework — baseline propagation logic refactored</td><td></td></tr>
<tr><td></td><td>Authentication with JWT + bcrypt completed. RBAC roles (Admin / User / Viewer) integrated with frontend tab-level access. Profile and password change pages delivered. INR tariff utility standardised. Full dashboard functional.</td><td>Final testing, validation, and documentation to be completed before second review submission</td><td>Cost currency inconsistency across modules resolved by centralised tariff utility</td><td></td></tr>
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Summary record of major progress meetings with supervisors (SPRINT 3)

Summary record of major progress meetings with supervisors			**Working title of dissertation/research project:** AI-Driven Smart Energy Optimization and Carbon Analytics Platform **.**	
Meeting date & supervisors present	**Progress since last meeting**	**Agreed programme of work and target dates**	**Other issues, e.g. facilities, supervision, training needs, etc.**	**Date of next meeting**
To be updated				

Worksheet / Data collection / Observation etc